

Isokinetic Sampling Train M9096

Metric Calculations

Profile Version

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1. Introduction

- 1.1. In the United Kingdom and Europe most regulatory and enforcement bodies require the data from sampling to be reported at Standard Temperature and Pressure (STP). In Europe this is defined as 273K, 101.325kPa.
- 1.2. In addition there may be requirements to provide data on a dry gas basis, wet gas basis or with correction to reference oxygen or carbon dioxide levels. This will be dependent on the process and the licence for the site and these must be consulted prior to submitting data to regulatory bodies.
- 1.3. Variations in equipment may occur and it is important to ensure that the units are those specified or are converted to these used in this manual.
- 1.4. In Appendix 2 are some conversion factors to assist in these manipulations.
- 1.5. It should be noted that in the USA Standard Conditions of Temperature and Pressure are 68°F (20°C) and 760 mmHg and hence some equations in this booklet will vary from those found in American Handbooks etc.

2. Calculations Required for Isokinetic Testing

2.1. Absolute pressure of the stack gas:

$$P_s = P_b + P_{static} \quad \text{mbar} \quad \text{Eq 1}$$

$$\begin{aligned} P_s &= \text{absolute Stack Pressure (mbar)} \\ P_b &= \text{barometric pressure (mbar)} \\ P_{static} &= \text{static Pressure (mbar)} \end{aligned}$$

2.2. Molecular weight of dry gas stream

$$M_d = 44 \times \frac{\%CO_2}{100} + 32 \times \frac{\%O_2}{100} + 28 \times \frac{\left(\frac{CO(ppm)}{10^4} + \%N_2\right)}{100} \quad \text{Eq 2}$$

$$\begin{aligned} M_d &= \text{Dry molecular weight of the stack gas} \\ \%CO_2 &= \text{carbon dioxide content in gas as a percentage} \\ \%O_2 &= \text{oxygen content in gas as a percentage} \\ \%N_2 &= \text{nitrogen content in gas as a percentage} \\ CO(ppm) &= \text{carbon monoxide content in gas in ppm} \end{aligned}$$

44, 32 and 28 are the molecular weights of CO₂, O₂, CO and N₂ respectively
(Nitrogen and CO both have a nominal molecular weight of 28)

2.3. Molecular Weight of Stack Gas (wet)

$$M_s = M_d \times \frac{(100 - B_{ws})}{100} + \frac{18 \times B_{ws}}{100} \quad \text{Eq 3}$$

B_{ws} = Percentage of water vapour in the gas stream on a volume/volume basis

M_s = Molecular weight of the stack gas on a wet basis

18 is the molecular weights of H₂O

2.4. Water Vapour in the Stack

2.4.1. In using the M9096 in-stack filtration equipment it is assumed that the water vapour is typically <5% and, or the temperature of the stack gases are above their dew point. However for accurate measurements of particulates the moisture level must be determined either from CEMS (Continuous Emissions Monitoring Systems, in EU AMS Automatic Measurement Systems). Alternatively this can be determined using US EPA Method 4 or BS EN14790:2005.

2.4.2. The Moisture Content as measured using volumes of gas and the volume of water present in the gas stream as a vapour

$$B_{ws} = \frac{V_{wc}}{V_{mc} + V_{wc}} \times 100 \quad \% \quad \text{Eq. 4}$$

$$V_{wc} = 0.001245 \times V_{H_2O} \quad m^3 \quad \text{Eq. 5}$$

B_{ws} = Percentage of water vapour present on a volume/volume basis

V_{mc} = Volume of Gas Pulled through the meter at STP (m³)

V_{wc} = Volume of Water as a Gas in the Stack Gases when V_{mc} (m³) of gas is pulled through the meter (m³)

V_{H_2O} = Volume of Condensed water collected in a Moisture Test (ml or g)

0.001245 = 22.414/(18 × 1000) (Molar Gas Volume at STP and molecular weight of water (18))

2.5. Calculation of the Ideal Nozzle Size for the Sample Train

$$D_n = \left(\left(\frac{\text{const.} Q_m P_m}{T_m C_p} \left(\frac{100 - B_{wm}}{100 - B_{ws}} \right) \right) \frac{T_s M_s}{P_s (\Delta P)_{avg}} \right) \quad (\text{mm}) \quad \text{Eq 6}$$

D_n = nozzle diameter (mm)

Q_m = ideal flow rate through the sample train ($l \text{ min}^{-1}$)

(typically between 5 and 25 $l \text{ min}^{-1}$ determined by the pump capacity and restrictions in the sample system such as orifice plates, silica gel etc.)

P_m = Pressure in the gas meter (in simple systems this is the barometric pressure P_b (mbar))

T_m = Absolute Temperature of the gas meter (in simple systems this is the ambient air temperature close to the meter ($T_{amb} ^\circ\text{C} + 273$) K

B_{wm} = Percentage of water vapour in the gas meter on a volume/volume basis, assumed to be zero as most gas meters are after the drying system %

B_{ws} = Percentage of water vapour in the stack gases on a volume/volume basis, %

C_p = Pitot tube coefficient (S-type = 0.84; Ellipsoidal type = 1), ideally should be a calibrated tip and this value used.

T_s = Absolute Temperature of the stack gases (Average Stack temp. $^\circ\text{C} + 273$) K

DP = Differential pressure of the velocity head as measured by the pitot tube (mbar)

(in calculating the ideal nozzle size it is usual to determine the differential pressure at a number of points according to the standards being followed and then to take the average of the square roots of these. Then to square this value and use this to determine the ideal nozzle size)

i.e. $(DP)_{avg} = (\text{average } (\sqrt{P_1} + \sqrt{P_2} + \sqrt{P_3} + \dots \sqrt{P_n} .)) ^2$ in many cases the simple average will suffice.

Const = 0.1646 (true for units specified)

2.6. Volume of Gas Sampled at Standard Conditions

2.6.1. At 273K, 101.325kPa the Corrected Volume of gas drawn by the gas meter will be:

$$V_{mc} = \frac{0.2695 \times V_m \times P_b \times Y_d}{T_m} \text{ m}^3 \quad \text{Eq. 7}$$

V_{mc} = Volume of Gas Pulled through the meter at STP and on a dry gas basis (m^3)

V_m = Volume of Gas Pulled through the meter as recorded by the meter (m^3)

Y_d = Gas Meter Calibration Factor to account for calibration of meter usually $Y_d = 1$

0.2695 = Correction factor to STP = 273/1013

2.7. Velocity of Stack Gases.

$$V_s = C_p \times \sqrt{\frac{2 \times \Delta P_{avg}}{\rho}} \quad \text{Eq. 8}$$

$$\rho = \frac{P}{R_{spec} \times T}$$

r = density of gas at Pressure P (Pa) and Temp T (K)

$$R_{spec} = \frac{R}{M} \quad \text{Eq. 9}$$

R = Boltzmann Gas Constant $8.31432 \times 10^3 \text{ Nm kmol}^{-1} \cdot \text{K}^{-1}$

M = Molecular weight of the specific gases

2.7.1. It Follows that:

$$V_s = C_p \times \sqrt{\frac{2 \times \Delta P_{avg}(Pa) \times Ts(K) \times R}{Ps(Pa) \times Ms}} \text{ m/s}$$

$$V_s = C_p \times \sqrt{\frac{2 \times \Delta P_{avg}(\text{mbar}) \times T_s(\text{K}) \times 8.3144 \times 10^3}{P_s(\text{mbar}) \times M_s}} \text{ m s}^{-1}$$

$$V_s = 128.95 \times C_p \times \sqrt{\frac{\Delta P_{avg}(\text{mbar}) \times T_s(\text{K})}{P_s(\text{mbar}) \times M_s}} \text{ m s}^{-1} \quad \text{Eq. 10}$$

C_p = Pitot tube coefficient (S-type = 0.84; Ellipsoidal type = 1), ideally should be a calibrated tip and this value used.
 M_s = Molecular weight of the stack gas on a wet basis

T_s = Absolute Temperature of the stack gases (Average Stack Temp. °C +273) K

DP = Differential pressure of the velocity head as measured by the pitot tube (mbar)

P_s = absolute Stack Pressure (mbar)

128.95 = Constant for Data conversion factors and R

2.8. Flow of Gases in the Duct

$$Q_a = 60 \times A_s \times V_s \quad \text{m}^3 \text{ min}^{-1} \quad \text{Eq. 11}$$

Q_a = Actual Flow of gases in Duct at Stack Gas Conditions of Temperature and Pressure (m³)
 A_s = Cross-sectional Area of the Stack (m²)
 60 = Conversion of seconds to minutes

2.9. Flow at Dry Standard Conditions

$$Q_{std} = \frac{0.002695 \times Q_a \times P_s \times (100 - B_{ws})}{T_s} \quad \text{m}^3 \text{ min}^{-1} \quad \text{Eq. 12}$$

2.10. Particulate Concentration at STP and on a dry gas basis

$$C_{std} = \frac{D_s}{V_{mc}} \quad \text{mg m}^{-3} \quad \text{Eq. 13}$$

C_{std} = Measured Concentration in the Stack Gases at STP and on a dry gas basis (mg Nm⁻³)
 D_s = Total amount of Dust Collected including washings (mg)
 V_{mc} = Volume of Gas Pulled through the meter at STP and on a dry gas basis (m³)

2.11. Particulate Concentration at STP and dry gas basis at reference Oxygen Conditions

$$C_{std\ O_{2ref}} = C_{std} \times \frac{(20.9 - O_{2ref})}{(20.9 - O_{2meas.})} \text{ mg m}^{-3} \quad \text{Eq. 14}$$

2.12. Particulate Emission Rate

$$E_s = \frac{C_{std} \times Q_{std} \times 60}{10^6} \quad \text{kg hr}^{-1} \quad \text{Eq. 15}$$

2.13. Percentage Isokinicity

2.13.1. When Isokineticity is achieved $V_s = V_n$ i.e Velocity in the stack is equal to the velocity entering the nozzle. Isokinicity is the measure of how close to this is achieved during the sampling.

$$I (\%) = \frac{V_n}{V_s} \times 100$$

2.13.2. The velocity in the stack is equal to Equation 10:

$$V_s = 128.95 C_p \sqrt{\frac{\Delta P_{avg} T_s M_s}{P_s}} \quad \text{m s}^{-1}$$

2.13.3. The flow entering the nozzle is determined from the flow rate through the Gas Meter. This is:

$$V_{mc} = \frac{0.2695 V_m P_b Y_d}{T_m} \text{ m}^3$$

2.13.4. The flow rate through the meter is:

$$Q_{mc.wet} = \frac{V_{mc}}{\theta}$$

θ = Time Pump was running and gas being sampled (min)

2.13.5. We need to ensure these are at the same conditions and hence the V_s is on a wet gas basis and the Volume from the meter is on a dry gas basis and needs to be at stack conditions of T_s & P_s

$$Q_{mc.wet} = \frac{V_{mc} \left(\frac{100}{100 - B_{ws}} \right)}{\theta} \times \frac{T_s}{273} \times \frac{1013}{P_s} \text{ m}^3 \text{ min}^{-1}$$

2.13.6. The velocity through the nozzle is

$$V_n = \frac{Q_{mc.wet}}{\text{Area of Nozzle}}$$

$$= \frac{V_{mc} \left(\frac{100}{100-B_{ws}} \right) \times 10^6 \times T_s \times 1013}{\Theta \times A_n \times 273 \times P_s \times 60} \text{ m s}^{-1}$$

2.13.7. So $I (\%) = \frac{V_n}{V_s} \times 100$

$$= \frac{V_{mc} \left(\frac{100}{100-B_{ws}} \right) \times 10^6 \times T_s \times 1013}{V_s \times \Theta \times A_n \times 273 \times P_s \times 60} \times 100$$

$$= \frac{6.1844 \times V_{mc} \times 10^6 \times T_s}{V_s \times \Theta \times A_n \times P_s \times (100-B_{ws})} \times 100$$

Finally

$$I (\%) = \frac{4 \times 6.1844 \times V_{mc} \times 10^6 \times T_s}{V_s \times \pi \times \Theta \times D_n^2 \times P_s \times (100-B_{ws})} \times 100 \quad \text{Eq. 16}$$

3. Setting the Flow Rate of the Sampler

3.1. The flow setting criteria is the same as determining the Isokinicity. We need to set $V_s = V_n$, i.e Velocity in the stack is equal to the velocity entering the nozzle.

3.2. When the above condition is met the flow rate is

$$Q_{m.wet \text{ at stack } T \& P} = V_n \times A_n$$

3.3. This needs to be corrected to gas meter conditions

$$\begin{aligned} Q_{m \text{ at Gas Meter Cond}} &= V_n \times A_n \times \left(\frac{100-B_{ws}}{100} \right) \times \left(\frac{T_m}{T_s} \right) \times \left(\frac{P_s}{P_m} \right) \\ &= 128.95 C_p \sqrt{\frac{\Delta P_{avg} T_s}{P_s M_s}} \left(\frac{\pi d_n^2}{4 \times 10^6} \right) \left(\frac{100-B_{ws}}{100} \right) \left(\frac{T_m}{T_s} \right) \left(\frac{P_s}{P_m} \right) \\ &= 128.95 \times C_p \sqrt{\frac{\Delta P_{avg} \cdot P_s}{T_s M_s}} \left(\frac{\pi d_n^2}{4 \times 10^6} \right) \left(\frac{100-B_{ws}}{100} \right) \left(\frac{T_m}{P_m} \right) \times 60 \times 10^3 \text{ l min}^{-1} \quad \text{Eq. 17} \end{aligned}$$

3.4. This is the flow that needs to be set on the sample unit to achieve Isokinicity.

3.5. For the M9096 unit Approximations are:

P_m	=	P_b barometric Pressure (mbar)
T_m	=	T_{amb} Absolute ambient air Temperature Close to Gas Meter (K)
M_s	=	28.86 if no CEMS available to determine molecular weight of gases

Appendix 1

Summary of Variables

Used In Calculations

P_s	=	absolute Stack Pressure (mbar)
P_b	=	barometric pressure (mbar)
P_{static}	=	static Pressure (mbar)
M_d	=	Dry molecular weight of the stack gas
%CO ₂	=	carbon dioxide content in gas as a percentage
%O ₂	=	oxygen content in gas as a percentage
%N ₂	=	nitrogen content in gas as a percentage
CO _(ppm)	=	carbon monoxide content in gas in ppm
B_{ws}	=	Percentage of water vapour present on a volume/volume basis
B_{ws}	=	Percentage of water vapour in the gas stream on a volume/volume basis
M_s	=	Molecular weight of the stack gas on a wet basis
V_{mc}	=	Volume of Gas Pulled through the meter at STP (m³)
V_{wc}	=	Volume of Water as a Gas condensed from the Stack Gases when V_{mc} (m³) of gas is pulled through the meter (m³)
V_{H_2O}	=	Volume of Condensed water collected in a Moisture Test (ml or g)
0.001245	=	22.414/(18 x 1000) (Molar Gas Volume at STP and molecular weight of water (18))
D_n	=	nozzle diameter (mm)
Q_m	=	ideal flow rate through the sample train (l min⁻¹)
P_m	=	Pressure in the gas meter (in simple systems this is the barometric pressure P_b (mbar))
T_m	=	Absolute Temperature of the gas meter (in simple systems this is the ambient air temperature close to the meter (T_{amb} °C+273) K)
B_{wm}	=	Percentage of water vapour in the gas meter on a volume/volume basis, assumed to be zero as most gas meters are after the drying system %
C_p	=	Pitot tube coefficient (S-type = 0.84; Ellipsoidal type = 1), ideally should be a calibrated tip and this value used.
T_s	=	Absolute Temperature of the stack gases (Average Stack temp. °C +273) K
DP	=	Differential pressure of the velocity head as measured by the pitot tube (mbar)
A_s	=	Cross-sectional Area of the Stack (m²)

i.e. $(DP)_{avg} = (\text{average } (\sqrt{P_1} + \sqrt{P_2} + \sqrt{P_3} + \text{etc.}))^2$ in many cases the simple average will suffice.

V_{mc}	=	Volume of Gas Pulled through the meter at STP and on a dry gas basis (m³)
V_m	=	Volume of Gas Pulled through the meter as recorded by the meter (m³)
Y_d	=	Gas Meter Calibration Factor to account for calibration of meter usually $Y_d = 1$
0.2695	=	Correction factor to STP = 273/1013
r	=	density of gas at Pressure P (Pa) and Temp T (K)
R	=	Boltzmann Gas Constant $8.31432 \times 10^3 \text{ Nm kmol}^{-1} \cdot \text{K}^{-1}$
M	=	Molecular weight of the specific gases
Q_a	=	Actual Flow of gases in Duct at Stack Gas Conditions of Temperature and Pressure (m³)
Q_{std}	=	Actual Flow of gases in Duct at STP (m³)
C_{std}	=	Measured Concentration in the Stack Gases at STP and on a dry gas basis (mg Nm⁻³)
D_s	=	Total amount of Dust Collected including washings (mg)
E_s	=	Emission Rate from the stack (kg hr⁻¹)
θ	=	Time Pump was running and gas being sampled (min)
$I(\%)$	=	Isokinicity(%)
Q_m	=	Flow through the meter at meter conditions (l min⁻¹)

Appendix 2

Useful Conversions of Units

Pressure

Pressure is measured in a variety of unit and as a result often leads to confusion, the following are useful inter-conversions:

1 mbar	=	10.2 mmH ₂ O
1 mbar	=	1 hPa
1 mbar	=	0.75006 mmHg
1 mbar	=	100 Pa
1013.25 mbar	=	760 mmHg
13.6 mmH ₂ O	=	1 mmHg
1 mmH ₂ O	=	0.09807 mbar
10332.3 mmH ₂ O	=	760 mmHg
1 mmH ₂ O	=	9.81 Pa
1 Pa	=	1 Nm ² (N is a Newton in this case not Normal as used elsewhere)
101325Pa	=	760 mmHg
1 mmHg	=	1.33322 mbar

Temperature

In the equations is always assumed to be in absolute units

Absolute Temp = Temperature in Celsius + 273 K

Appendix 3

Example Calculations

P_b	=	1029 (mbar)
P_{static}	=	+0.49 (mbar)
%CO ₂	=	1%
%O ₂	=	20%
%N ₂	=	79%
CO _(ppm)	=	0 ppm
B _{ws}	=	3%
D _n	=	To be Determined (mm)
Q _m	=	16.67 ideal flow rate through the sample train (l min ⁻¹)
P _m	=	1029 (in simple systems this is the barometric pressure P _b (mbar))
T _m	=	21°C + 273 = 294 K
B _{wm}	=	Percentage of water vapour in the gas meter on a volume/volume basis, assumed to be zero as most gas meters are after the drying system %
C _p	=	0.84
T _s	=	150°C+273= 423 K
DP	=	2.0196 (mbar)

Absolute pressure of the stack gas:

$$P_s = 1029 + 0.49 = 1029.49 \text{ mbar}$$

Molecular weight of dry gas stream

$$M_d = 44 \times \frac{1}{100} + 32 \times \frac{20}{100} + 28 \times \frac{\left(\frac{0}{10^4} + 79\right)}{100} = 28.96$$

Molecular Weight of Stack Gas (wet)

$$M_s = 28.96 \times \frac{(100-3)}{100} + \frac{18 \times 3}{100} = 28.63$$

Calculation of the Ideal Nozzle Size for the Sample Train

$$D_n = \left(\left(\frac{0.1646 \times 16.7 \times 1029}{294 \times 0.84} \left(\frac{100 - 0}{100 - 3} \right) \right) \frac{423 \times 28.63}{1029.49 \times 2.0196} \right) \text{ (mm)}$$

$$= 5.33 \text{ mm}$$

The ideal flow rate Q_m is 1 m^3 per hour which is 16.67 l min^{-1} in this example it can be any rate within the capacity of the system being used.

$P_m = P_b$ = Barometric Pressure and T_m is the ambient air temp in the vicinity of the gas meter

It is unlikely that a 5.33 mm nozzle would be available so it is usual practice to choose the nearest size available. It is safest to choose the slightly smaller nozzle than calculated as this will ensure that there is sufficient capacity to sample all the sample points. In this case we would choose the **5 mm** nozzle rather than the 6 mm nozzle also available. However this will also result in a slightly smaller sample of gas being obtained.

Setting the Flow Rate of the Sampler

The flow setting criteria is similar to determining the Isokinicity. We need to set $V_s = V_n$, i.e Velocity in the stack is equal to the velocity entering the nozzle.

$$Q_{m @ \text{ Meter Cond}} = 128.95 \times 0.84 \sqrt{\frac{2.0196 \times 1029.49}{423 \times 28.63}} \left(\frac{3.142 \times 5^2}{4 \times 10^6} \right) \left(\frac{100 - 3}{100} \right) \left(\frac{294}{1029} \right) \times 60 \times 10^3 \text{ l min}^{-1}$$

$$= 14.65 \text{ l min}^{-1}$$

Determination of Particulate Sampling Results

V_{H_2O}	=	25 (ml or g)
D_n	=	5.0 (mm)
P_m	=	1029 P_b (mbar)
T_m	=	19°C + 273 (K)
C_p	=	Pitot tube coefficient (S-type = 0.84)
T_s	=	142 °C + 273 (K)
DP	=	2,353 (mbar)
A_s	=	0.785 (m ²) Cross-sectional Area of the Stack
V_m	=	0.950 (m ³) Volume of Gas Pulled through the meter as recorded by the meter
Y_d	=	1.0013 Gas Meter Calibration Factor to account for calibration of meter
θ	=	60 (min) Time Pump was running and gas being sampled

$$D_s = 45 \text{ (mg)} \text{ Total amount of Dust Collected including washings}$$

Water Vapour in the Stack

$$V_{wc} = 0.001245 \times 25 \text{ m}^3$$

$$= 0.0311 \text{ m}^3$$

Volume of Gas Sampled at Standard Conditions

At 273K, 101.325kPa the Corrected Volume of gas drawn by the gas meter will be:

$$V_{mc} = \frac{0.2695 \times 0.950 \times 1029 \times 1.0013}{292} \text{ m}^3$$

$$= 0.903 \text{ m}^3 \text{ Volume of Gas Pulled through the meter at STP and on a dry gas basis (m}^3\text{)}$$

$$B_{ws} = \frac{0.0311}{0.903 + 0.0311} \times 100 \%$$

$$= 3.3\%$$

Velocity of Stack Gases.

$$V_s = 128.95 \times 0.84 \times \sqrt{\frac{2.353 \times 415}{1029.49 \times 28.63}} \text{ m s}^{-1}$$

$$= 19.7 \text{ m s}^{-1}$$

Flow of Gases in the Duct

$$Q_a = 60 \times 0.785 \times 19.7 \text{ m}^3 \text{ min}^{-1}$$

$$= 927.87 \text{ m}^3 \text{ min}^{-1}$$

Flow at Dry Standard Conditions

$$\begin{aligned} Q_{std} &= \frac{0.002695 \times 927.87 \times 1029.49 \times (100-3.3)}{415} \quad \text{m}^3 \text{ min}^{-1} \\ &= 599.85 \text{ m}^3 \text{ min}^{-1} \end{aligned}$$

Particulate Concentration at STP and on a dry gas basis

$$\begin{aligned} C_{std} &= \frac{45}{0.903} \quad \text{mg m}^{-3} \\ &= 49.8 \text{ mg m}^{-3} \end{aligned} \quad \begin{array}{l} \text{Particulate Concentration at 273K,} \\ \text{101.325kPa and on a dry gas basis} \end{array}$$

3.6. Particulate Emission Rate

$$\begin{aligned} E_s &= \frac{49.8 \times 599.85 \times 60}{10^6} \quad \text{kg hr}^{-1} \\ &= 1.79 \text{ kg hr}^{-1} \end{aligned}$$

Particulate Concentration at STP and on a dry gas basis at 15% O₂

$$\begin{aligned} C_{std \text{ O}_{2ref}} &= 49.8 \times \frac{(20.9-15)}{(20.9-20)} \quad \text{mg m}^{-3} \\ C_{std} &= 326 \text{ mg m}^{-3} \end{aligned}$$

Percentage Isokinicity

$$\begin{aligned} I (\%) &= \frac{4 \times 6.1844 \times 0.903 \times 10^6 \times 415}{19.7 \times 3.142 \times 60 \times 5.02 \times 1029.49 \times (100-3.3)} \times 100 \\ &= 100.3 \% \end{aligned}$$